# The first law

**antocciphysics**



**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Use a consistent sign convention in an energy balance.

**SUCCESS CRITERIA - I can...**

1. State whether heat enters or leaves.
2. Distinguish work done by the system from work done on it.
3. Apply ΔU=Q-W with the stated signs.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Bring back an earlier idea

A 3 kg sample completely changes phase with latent heat 500 J/kg. Find the energy magnitude required.

**Marzano check - circle:** ① ② ③

## 02 Explain a worked example

**Reference:** $\Delta U = Q - W_{by}$.
Q is positive into the system; $W_{by}$ is positive when the system does work on its surroundings.

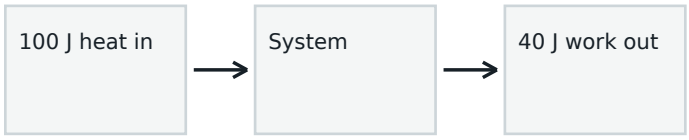
**Worked reasoning:** A system receives 90 J of heat and does 30 J of work on its surroundings. With ΔU=Q-W, internal energy increases by 60 J.

**Your turn:** Draw labeled arrows crossing a system boundary and mark the 60 J stored increase.

**Marzano check - circle:** ① ② ③

## 03 Look and explain

A system absorbs 100 J of heat and does 40 J of work. Draw the energy arrows and find its internal-energy change.




**Marzano check - circle:** ① ② ③

## 04 Build the solution

A system receives 80 J and does 20 J of work. Complete ΔU=80-20=____ J. Does internal energy increase or decrease?

**Marzano check - circle:** ① ② ③

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### 05 Use the idea

Why does the phrase “work is 60 J” need a direction or sign convention?

**Marzano check - circle:** 1 2 3

### 06 Choose a method

Total kinetic energy changes from 24 J to 20 J in an isolated collision. Is it elastic? Where could the difference go?

**Marzano check - circle:** 1 2 3

### 07 Exit check

Heat input 30 J; work output 10 J. Find ΔU.

**Marzano check - circle:** 1 2 3

### 08 STRETCH: Apply it another way

Give two different heat-input/work-output pairs that increase internal energy by 40 J.

**Marzano check - circle:** 1 2 3